

## Investigations on dry sliding wear behaviour of nylon-6 and GFR nylon-6 composites using Taguchi technique

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Polymer materials have replaced the metals due to intrinsic lubrication property, superior wear resistance, excellent strength and stiffness to weight ratio, chemical resistance, corrosion resistance and low processing cost. In food and chemical industry polymers are widely used in gear and other sliding machining elements because excellent chemical resistance and good lubrication properties. So In this paper, tribological properties of nylon-6 and glass fiber reinforced (GFR) nylon 6 composites are investigated. Nylon-6 and GFR nylon-6 are injection molded for the fabrication of pins and tested for coefficient of friction and specific wear rate against steel disc under dry condition on a pin-on-disc configuration. The coefficient of friction and specific wear rate are studied at varying glass fiber content (0, 10, 20 and 30 wt%), different sliding velocity (0.5, 1.0, 1.5 and 2.0 m/s), applied load (5, 10, 15 and 20 N), and constant sliding distance (1000 m) under atmospheric condition of temperature and humidity. The disc specimen is made of AISI D2 steel. Process parameters are optimized by taguchi technique. Scanning electron microscope is used further for the study of wear mechanism of worn surface of specimens.

**Keywords:** Polymer, Steel, Mechanical properties, Tribological properties, Taguchi technique

Friction, wear, fatigue and corrosion are the major drawbacks found by industries which lead to replacement of parts. In recent years, the tribological applications of polymer materials have been broadly used in sliding parts in food and chemical industries such as gears and cams. This is because polymers have intrinsic lubrication behavior which avoids the use of oil lubrication. Adhesion and deformation mechanisms are mainly involved in wear and friction of polymer. The adhesion mechanism of friction of polymer describes breakage of weak bonds between polymer pin and steel disc. The deformation mechanism describes the dissipation of energy in the contact area<sup>1</sup>. Polymers showed low frictional coefficient comparison to metal because of their less interfacial adhesion energy<sup>2</sup>. Polymers have low wear resistance, mechanical strength and low thermal conductivity so that several reinforcements and filler materials are mixed to the polymer to upgrade their tribological, mechanical and thermal behaviour<sup>3,4</sup>. Nylon has superior wear resistance, intrinsic lubrication behavior and good tensile strength owing to hydrogen bond and van der Waals force present in molecular chains of nylon<sup>5</sup>. Wear mainly depends on the cohesion and adhesion of the transfer film to the disc surface and

protection of metal disc by transfer layer. The main cause of wear loss of material was observed before transfer layer formed on disc and loss of transfer layer by peeling from disc<sup>6,7</sup>. Eyre<sup>8</sup> studied that wear of materials encountered in industrial situations are abrasive wear 50%, adhesive wear 15%, erosion wear 8%, fretting wear 8%, chemical wear 5% and others contribution are 14% as shown in Fig. 1. Friedrich and Schlarb<sup>9</sup> concluded that the friction and wear behavior of polymer under various experimental conditions

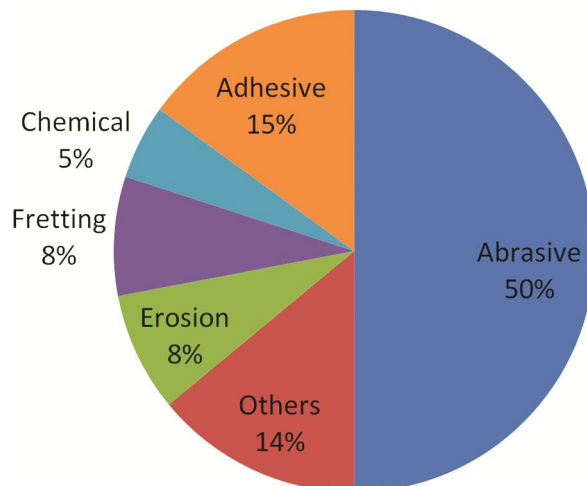


Fig. 1 — Type of wear in industry (approximate percentage involved)<sup>8</sup>

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depends upon the properties like crystallinity, glass transition temperature, mechanical properties, molecular weight, orientation, hardness and surface energy. On the other hand, the tribological system itself, more precisely the loading characteristics, the counterpart material, as well as the external conditions including the temperature or the presence of lubricants play a major role in wear mechanism and subsequently for the overall wear performance. An overview of the various factors influencing the wear behavior of polymers is shown in Fig. 2.

The friction coefficient and specific wear rate of polyamide 12 composites decreased with increasing glass fiber and best obtained with the composites containing 30 wt. % glass fibers. Further if glass fiber increased then friction coefficient and specific wear rate increased<sup>10</sup>. If glass fiber content increased the wear resistance improved and friction coefficient decreased. They observed that fiber reinforced showed more wear resistance against steel disc due to transfer film formed on steel disc<sup>11,12</sup>. By using glass and carbon fiber in nylon matrix, both the tribological and mechanical properties improved. When glass or aramid fiber is used the wear resistance improves when sliding against the steel disc. However, the behavior is mainly affected by many factors like type, shape, size and reinforcement of the fibers, matrix materials used and the test condition in which the experiment is conducted<sup>13</sup>. The applied load and sliding velocity parameter was more effective on

coefficient of friction and specific wear rate. If applied load increases the coefficient of friction decreases but only limited at elastic constant, and when plastic deformation starts then friction coefficient slightly increases due to further increase in load<sup>14</sup>. The coefficient of friction and specific wear rate of glass fiber reinforced nylon-6 composites was lowest at GF30%<sup>15</sup>. Xing *et al.*<sup>16,17</sup> found that if sliding velocity increased coefficient of friction decreased due to change in the temperature of surface of the specimen.

The objective of this paper is to study specific wear rate, coefficient of friction and wear mechanism of nylon-6 and GFR nylon-6 materials against steel disc. The Glass wt%, applied load, sliding velocity was varied and the sliding distance of 1000 m was maintained constant throughout the experiment. Worn surface morphology of pin and disc specimens were also discussed.

## Experimental Procedure

### Materials

The materials used in the study were nylon-6 and GFR nylon-6 in form of granules as shown in Fig. 3. The pin with various amount of GFR nylon-6 (0, 10, 20, and 30 wt%) was produced by injection moulding machine (Modern Plastic and Equipment's, Model-MPE-TLH-01). The tensile properties of the nylon-6 and GFR nylon-6 was evaluated using ASTM D638-

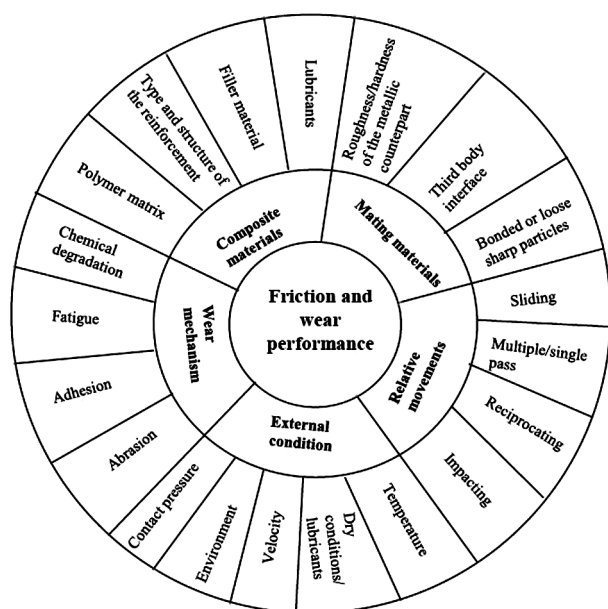


Fig. 2 — Factors determining the wear and friction behavior of polymers

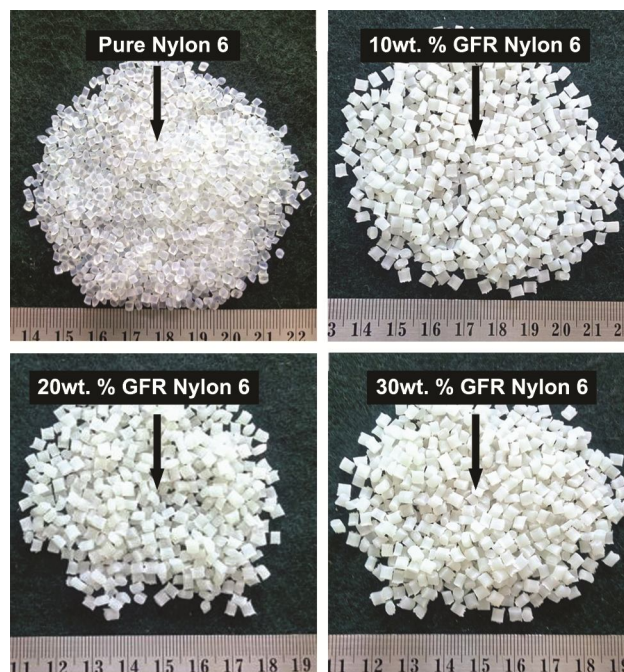


Fig. 3 — Pure nylon-6, 10, 20 and 30 wt% glass fiber reinforced nylon-6 (granules form)

14, Shore D hardness of the pin specimen was evaluated using ASTM D2240-05 and notched impact strength was evaluated using ASTM D 256-10 as given in Table 1. The temperature of injection molding machine is maintained at 220°C, 225°C, 230°C and 240°C for 0, 10, 20 and 30 wt% GFR nylon-6, respectively. The dimension of pin specimen was 31 mm length and 6 mm diameter and steel disc specimen was 70 mm diameter and 7mm thick. The Chemical composition of AISI D2 steel disc is given in Table 2.

#### Friction and wear testing

The tribological test was conducted under dry condition as per ASTM G-99-05 standard on (DUCOM TR-20M-106) pin-on-disc arrangement with nylon-6 and GFR nylon-6 pin specimens sliding against AISI D2 steel disc specimens. Figure 4 shows the schematic diagram of pin-on-disc set-up. The friction and wear tests were conducted at various glass fiber contents (0, 10, 20 and 30 wt%) applied load (5, 10, 15 and 20 N), sliding velocity (0.5, 1.0, 1.5 and 2.0 m/s) and constant

sliding distance (1000 m) under atmospheric condition of temperature (23°C) and humidity (67 ±10). Density of nylon-6 was observed 0.00113 g/mm<sup>3</sup>, 10 wt% GFR nylon-6 was 0.00120 g/mm<sup>3</sup>, 20 wt% GFR nylon-6 was 0.00127 g/mm<sup>3</sup> and 30 wt% GFR nylon-6 was 0.00135 g/mm<sup>3</sup>. Before testing, all the pin specimens were polished using a 600 grade SIC emery paper to ensure full contact with disc and achieved same roughness. Pin and disc specimen was cleaned using cotton soaked in acetone and dried before the test. The initial weight before experiment and final weight after experiment of specimen were weighted using an electronic digital analytical balance having an accuracy of 0.1 mg.

The specific wear rate  $K_s$  (mm<sup>3</sup>/Nm) was calculated by using Eq. (1).

$$\text{Specific wear rate}(K_s) = \frac{m_1 - m_2}{\rho \times N \times S} \quad \dots(1)$$

Where  $m_1$  and  $m_2$  are mass of the pin specimen before and after experiment (g),  $\rho$  represents density of the pin specimen (g/mm<sup>3</sup>),  $N$  represents applied load (N), and  $S$  represents sliding distance (m).

Table 1 — Mechanical properties of pure and GFR nylon-6 composites

| Sl. No. | Glass fiber content (wt%) | Shore D hardness ASTM D 2240 | Notched impact strength(J/mm) ASTM D256 | Ultimate tensile strength (N/mm <sup>2</sup> ) ASTM D 638 |
|---------|---------------------------|------------------------------|-----------------------------------------|-----------------------------------------------------------|
| 1       | 0%                        | 63-65                        | 0.137                                   | 51.208                                                    |
| 2       | 10%                       | 63-67                        | 0.075                                   | 53.631                                                    |
| 3       | 20%                       | 72-74                        | 0.125                                   | 71.358                                                    |
| 4       | 30%                       | 73-75                        | 0.125                                   | 86.014                                                    |

Table 2 — Chemical Composition of AISI D2 Steel

| Element           | Fe     | Cr    | Mn    | Mo   | P     | Si    | V     | S     | C    |
|-------------------|--------|-------|-------|------|-------|-------|-------|-------|------|
| Composition (wt%) | 84.881 | 11.52 | 0.156 | 0.72 | 0.018 | 0.302 | 0.927 | 0.006 | 1.47 |

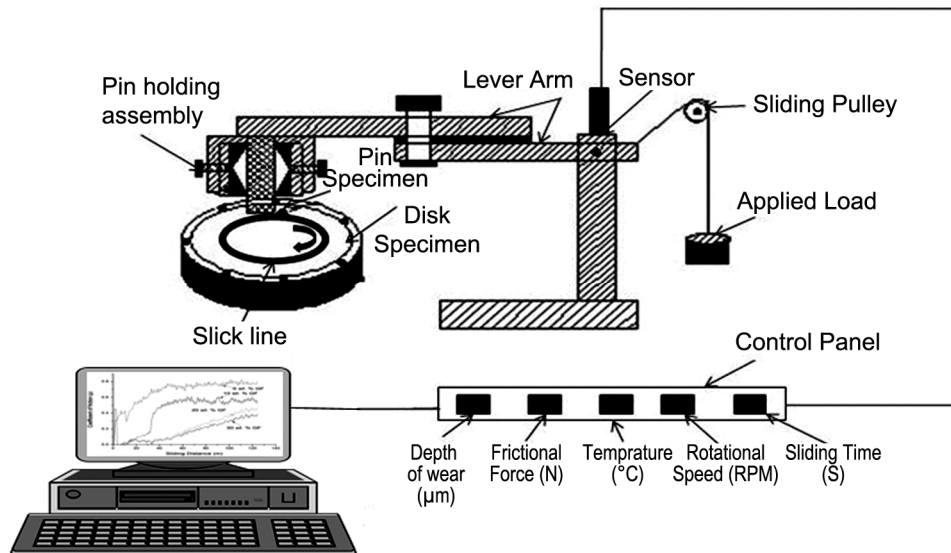


Fig. 4 — Schematic diagram of pin-on-disc setup

### Taguchi's design of experiment

Taguchi is a powerful tool for optimizing the performance through the settings of design parameters and decreasing the fluctuation of system performance. Minitab statistical software can be used to explain the parameter effect. Taguchi method can be used to determine both the optimal result from finite analytical data and the dominant factors involved in the optimization for friction and wear. Taguchi method uses a special design of orthogonal arrays to study the entire parameter space with a small number of experiments only. The Taguchi method uses the signal to noise (S/N) ratio. The experimental results are transformed into an S/N ratio. Usually, there are three categories of quality characteristics in the analysis of the S/N ratio, i.e., lower-the-better, higher-the-better, and nominal-the-better. Here smaller the better characteristic is used as shown in Eq. (2).

$$S/N = -10 \log \left[ \frac{1}{n} \sum_{i=1}^n y_i^2 \right] \quad \dots (2)$$

Where  $y$  represents average observed data and  $n$  represents number of tests.

The coefficient of friction and specific wear rate was carried out with three control factors such as glass fiber contents, applied loads and sliding velocity each has four levels as shown in Table 3.

### Analysis of variance

Analysis of variance (ANOVA) is a statistical technique used to predict the significance of process parameters on the quality characteristics. Using Minitab 16 software, ANOVA is performed to determine parameter which has a significant effect on the performance<sup>18,19</sup>.

## Results and Discussion

### Effect of glass fiber content, applied load and sliding velocity on coefficient of friction and specific wear rate

The variation of coefficient of friction with respect to sliding distance at varying glass fiber content irrespective of applied loads is shown in Fig. 5. During sliding applied load shared by both glass fiber and nylons shown in Fig. 6, but major part of applied load was shared by glass fiber. When the glass fiber

content increases the coefficient of friction decreases. The coefficient of friction was highest in the case of pure nylon-6 and on further addition of glass fiber the coefficient of friction decreased as shown in Fig. 7. The major cause for variation in coefficient of friction is due to transfer layer produced on disc, decreased adhesion of the pure nylon-6 and temperature of contact zone. In pure nylon-6 cases the coefficient of friction was highest and in case of GFR nylon-6 the coefficient of friction is less due to formation of transfer layer on steel disc. If applied load increases then coefficient of friction decreases shown in Fig. 7. When sliding velocity increased coefficient of friction decreased due to change in temperature of surface of specimen. In case of glass fiber content due to rubbing of glass fiber of faced material the coefficient of friction decreased.

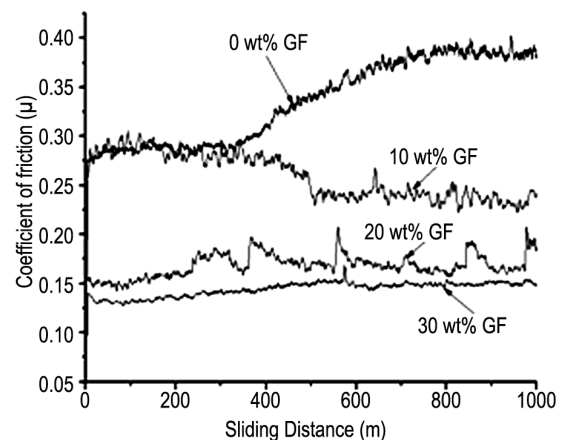


Fig. 5 — Relation between sliding distance and coefficient of friction under varying glass fiber content at velocity 0.5 m/s irrespective of load

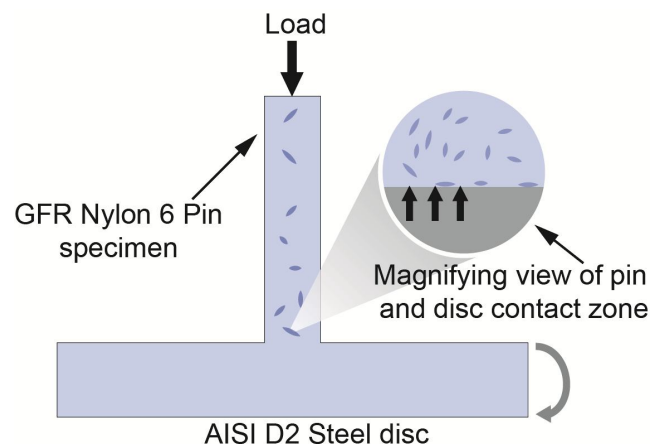


Fig. 6 — Schematic representation of load sharing between fibres and the matrix

Table 3 — Control factors and their levels

| Control factors |                       | Levels |     |     |     |
|-----------------|-----------------------|--------|-----|-----|-----|
|                 |                       | 1      | 2   | 3   | 4   |
| A               | Glass fiber (wt %)    | 0      | 10  | 20  | 30  |
| B               | Applied load (N)      | 5      | 10  | 15  | 20  |
| C               | Sliding velocity(m/s) | 0.5    | 1.0 | 1.5 | 2.0 |

Table 4 — Experimental results for coefficient of friction and specific wear rate

| Exp run | GFR (wt%) | Load (N) | Sliding velocity (m/s) | Coefficient of friction ( $\mu$ ) | Specific wear rate ( $\text{mm}^3/\text{Nm}$ ) |
|---------|-----------|----------|------------------------|-----------------------------------|------------------------------------------------|
| 1       | 0         | 5        | 0.5                    | 0.363                             | 0.0000176                                      |
| 2       | 0         | 10       | 1.0                    | 0.266                             | 0.0000176                                      |
| 3       | 0         | 15       | 1.5                    | 0.260                             | 0.0000294                                      |
| 4       | 0         | 20       | 2.0                    | 0.231                             | 0.0000442                                      |
| 5       | 10        | 5        | 1.0                    | 0.255                             | 0.0000166                                      |
| 6       | 10        | 10       | 0.5                    | 0.248                             | 0.0000166                                      |
| 7       | 10        | 15       | 2.0                    | 0.213                             | 0.0000389                                      |
| 8       | 10        | 20       | 1.5                    | 0.195                             | 0.0000333                                      |
| 9       | 20        | 5        | 1.5                    | 0.185                             | 0.0000314                                      |
| 10      | 20        | 10       | 2.0                    | 0.191                             | 0.0000315                                      |
| 11      | 20        | 15       | 0.5                    | 0.169                             | 0.0000104                                      |
| 12      | 20        | 20       | 1.0                    | 0.165                             | 0.0000157                                      |
| 13      | 30        | 5        | 2.0                    | 0.184                             | 0.0000296                                      |
| 14      | 30        | 10       | 1.5                    | 0.151                             | 0.0000222                                      |
| 15      | 30        | 15       | 1.0                    | 0.163                             | 0.0000197                                      |
| 16      | 30        | 20       | 0.5                    | 0.146                             | 0.0000111                                      |

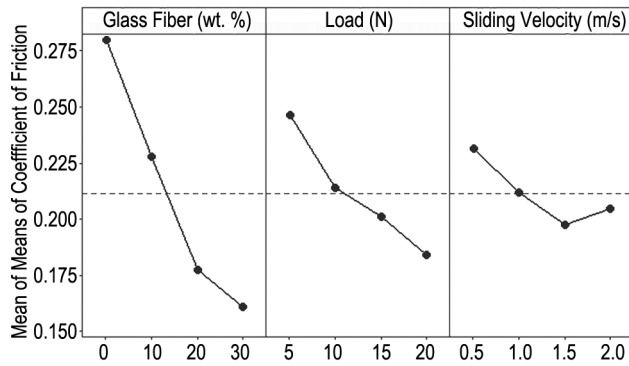


Fig. 7 — Effect of control factor on coefficient of friction for mean

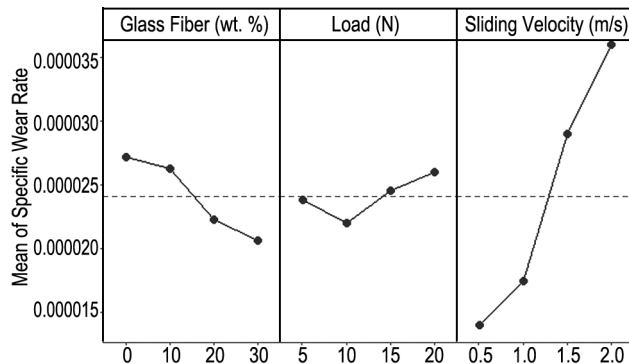


Fig. 8 — Effect of control factor on specific wear rate for mean

Specific wear rate was observed between  $1.049\text{E-}5 \text{ mm}^3/\text{Nm}$  to  $4.424\text{E-}5 \text{ mm}^3/\text{Nm}$  as shown in Table 4. In case of pure nylon-6, specific wear rate observed high and when glass fiber content increased then specific wear rate decreased as shown in Fig. 8. When glass fiber content increases hardness, tensile strength

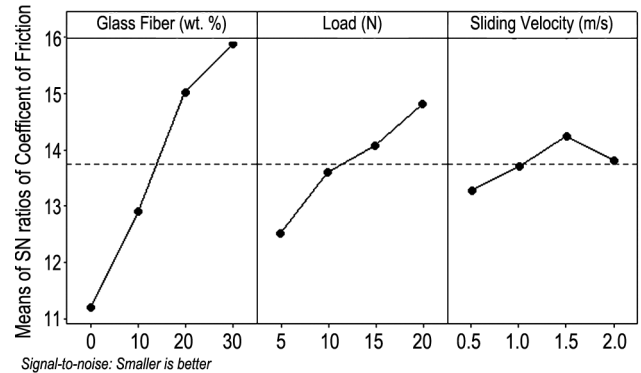


Fig. 9 — Effect of control factor on coefficient of friction for S/N ratio

and thermal conductivity increases so wear resistance improved. When applied load increases specific wear rate increases due to more heat produced at contact zone. The specific wear rate increases if sliding velocity increases due to increase in temperature of the contact zone.

#### Taguchi analysis and ANOVA

The aim of this study is to find the combination of factors to achieve the minimum coefficient of friction and specific wear rate. The combination of factors  $A_4B_4C_3$  offer minimum coefficient of friction Fig. 9. The combination of factors  $A_4B_2C_1$  offer minimum specific wear rate as shown in Fig. 10. Table 5 represents the S/N ratio response table for the coefficient of friction. The Delta value which is the deviation between maximum and minimum S/N ratios gives the effect of influencing factor on the

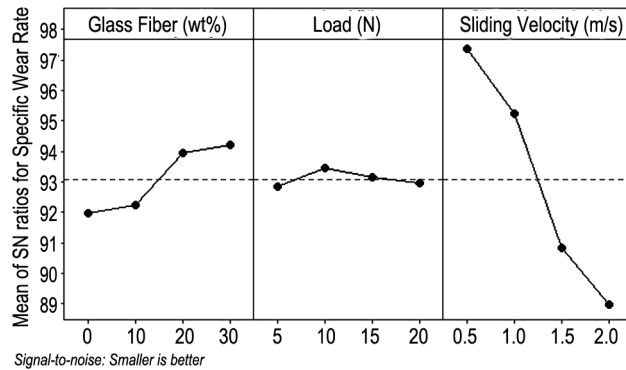


Fig. 10 — Effect of control factor on specific wear rate for S/N ratio

Table 5 — S/N ratio response Table for coefficient of friction

| Level | S/N Ratios  |              |                  |
|-------|-------------|--------------|------------------|
|       | Glass fiber | Applied load | Sliding velocity |
| 1     | 11.18       | 12.51        | 13.27            |
| 2     | 12.90       | 13.60        | 13.69            |
| 3     | 15.03       | 14.08        | 14.24            |
| 4     | 15.90       | 14.82        | 13.81            |
| Delta | 4.72        | 2.31         | 0.98             |
| Rank  | 1           | 2            | 3                |

Table 6 — S/N ratio response Table for specific wear rate

| Level | S/N Ratios  |              |                  |
|-------|-------------|--------------|------------------|
|       | Glass fiber | Applied load | Sliding velocity |
| 1     | 91.98       | 92.83        | 97.36            |
| 2     | 92.24       | 93.45        | 95.22            |
| 3     | 93.96       | 93.15        | 90.83            |
| 4     | 94.21       | 92.95        | 88.98            |
| Delta | 2.24        | 0.62         | 8.38             |
| Rank  | 2           | 3            | 1                |

Table 7 — Analysis of variance for coefficient of friction

| Source                 | DF | Seq SS    | Adj MS    | F-ratio | P     | % contribution |
|------------------------|----|-----------|-----------|---------|-------|----------------|
| Glass fiber            | 3  | 0.0346502 | 0.0115501 | 25.41   | 0.001 | 71.73          |
| Load (N)               | 3  | 0.0083857 | 0.0027952 | 6.15    | 0.029 | 17.36          |
| Sliding velocity (m/s) | 3  | 0.0025407 | 0.0008469 | 1.86    | 0.237 | 5.25           |
| Residual Error         | 6  | 0.0027274 | 0.0004546 | -       | -     | 5.64           |
| Total                  | 15 | 0.0483039 | -         | -       | -     | 100            |

Where, DF-Degree of freedom, Seq SS-Sequential sums of squares, Adj MS-Adjusted mean squares, P-Probability

Table 8 — Analysis of variance for Specific wear rate

| Source                 | DF | Seq SS    | Adj MS    | F     | P     | %contribution |
|------------------------|----|-----------|-----------|-------|-------|---------------|
| Glass fiber            | 3  | 0.0000058 | 0.0000019 | 12.21 | 0.006 | 31.69         |
| Load (N)               | 3  | 0.0000036 | 0.0000012 | 7.64  | 0.018 | 19.67         |
| Sliding velocity (m/s) | 3  | 0.0000078 | 0.0000026 | 16.44 | 0.003 | 42.62         |
| Residual Error         | 6  | 0.0000010 | 0.0005371 | -     | -     | 5.46          |
| Total                  | 15 | 0.0000183 | -         | -     | -     | 100           |

coefficient of friction. It is found out that glass fiber is having the most effect and it is followed by applied load and sliding velocity. Table 6 represents the S/N ratio response table for specific wear rate. It is found out that sliding velocity is having the most effect and it is followed by glass fiber and applied load. Table 7 shows the results of the ANOVA with the coefficient of friction. On the basis of ANOVA for minimum coefficient of friction observed that glass fiber content 71.73%, sliding velocity 5.25 % and applied load 17.36% exerts a significant influence. Table 8 gives the results of the ANOVA with specific wear rate wear. For specific wear rate the glass fiber content 31.69% has the most contribution on the output followed by sliding velocity 42.62% and applied load 19.67%<sup>3,4</sup>.

#### Confirmation tests for Taguchi analysis

Once the optimal level of the design parameters has been determined using Minitab software for coefficient of friction (0.146) and specific wear rate (0.0000109) are  $A_4B_4C_3$  and  $A_4B_2C_1$  respectively. The final step is to predict and verify the improvement of the quality characteristic using the optimal level of the design parameters. For the optimized design parameters experiment trails were carried out to estimate the experimental value of coefficient of friction (0.142) and specific wear rate (0.0000105). A very low % error of 2.73% and 3.66% was obtained for coefficient of friction and specific wear rate respectively between the predicted and experimental values showing a good correlation as shown in Table 9.

#### Wear mechanism

Figures 11(a-b) shows the mechanism of adhesive wear phenomena. Microscopy of the worn surfaces of pure nylon-6 at 20 N applied load shown in Fig. 12a.



| Table 9 — Confirmation test                                                         |                            |                    |                    |                                                                        |
|-------------------------------------------------------------------------------------|----------------------------|--------------------|--------------------|------------------------------------------------------------------------|
| Optimal parameters                                                                  |                            |                    |                    | Error (%) = ((Prediction value-Experiment value)/Prediction value)*100 |
| Parameter levels                                                                    | Prediction value           | Experiment value   |                    |                                                                        |
| Coefficient of friction ( $\mu$ )<br>Specific wear rate ( $\text{mm}^3/\text{Nm}$ ) | $A_4B_4C_3$<br>$A_4B_2C_1$ | 0.146<br>0.0000109 | 0.142<br>0.0000105 | 2.73<br>3.66                                                           |

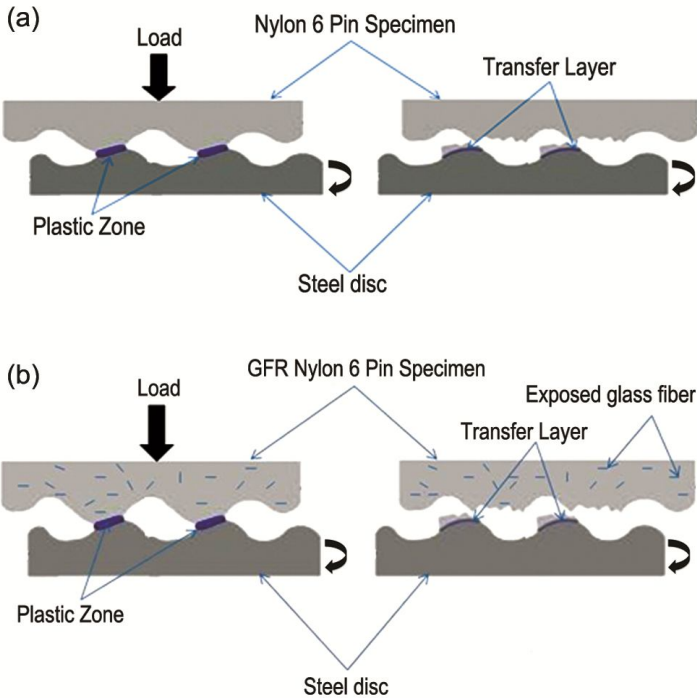


Fig. 11 — Mechanism of adhesive wear (a) nylon-6 and (b) GFR nylon-6

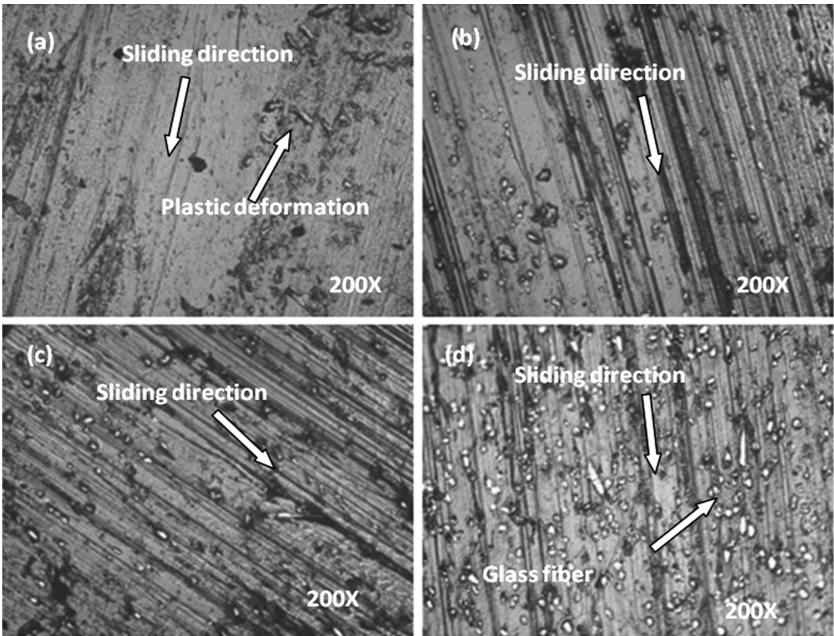


Fig.12 — Microscopy of worn surfaces of the nylon-6 composites under varying glass fiber content and velocity, at a load of 20 N : (a) 0 wt% GF, (b) 10 wt% GF, (c) 20 wt% GF and (d) 30 wt% GF irrespective of sliding velocity

In this case some adhesion with smearing and melting with sliding direction was observed. This is due to the temperature rise of the contact zone. Figures 12(b-d) show the worn out surface of glass fiber reinforced nylon-6 composites. This is the main cause to lower the specific wear rate. Figures 12(b-d) show that when glass fiber content increased then more rubbed glass fiber was observed on the surface of nylon-6 composites. Figures 12a and 13a show smearing and melting of polymer means plastic deformation due to less thermal conductivity of pure nylon. Figures 12d and 13b show debris patches with sliding direction here the glass fiber found on the sliding surface of nylon-6 composites; it played a significant role of sharing load. When GFR nylon-6 slides against steel disc, transfer films are observed. This transfer film produced a cover to nylon-6 composites from steel disc. In Fig. 14, worn surface of steel disc shows continuous transfer film over the wear track. Microscopy of the worn surfaces of pure nylon-6 at 5 N and 20 N applied load shown in

Fig. 15(a,b). At 5 N load worn surface showing wrinkle with sliding direction while at 20 N adhesion with smearing and melting with sliding direction was observed. This is due to the temperature rise of the contact zone.

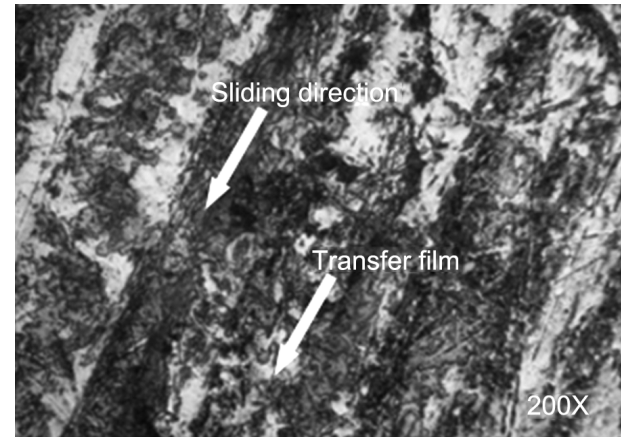


Fig. 14 — Transfer film formed on the steel disc surface testing under condition at 30 wt% GFR nylon-6 and 20 N loads irrespective of sliding velocity

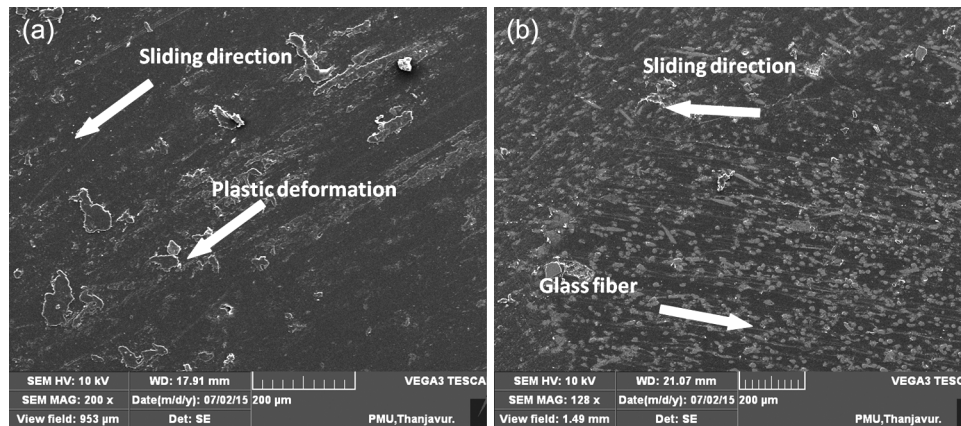


Fig. 13 — SEM micrographs of the worn surfaces of the nylon-6 composites under varying glass fiber content and velocity, at a load of 20 N : (a) 0 wt% GF and (b) 30 wt% GF irrespective of sliding velocity

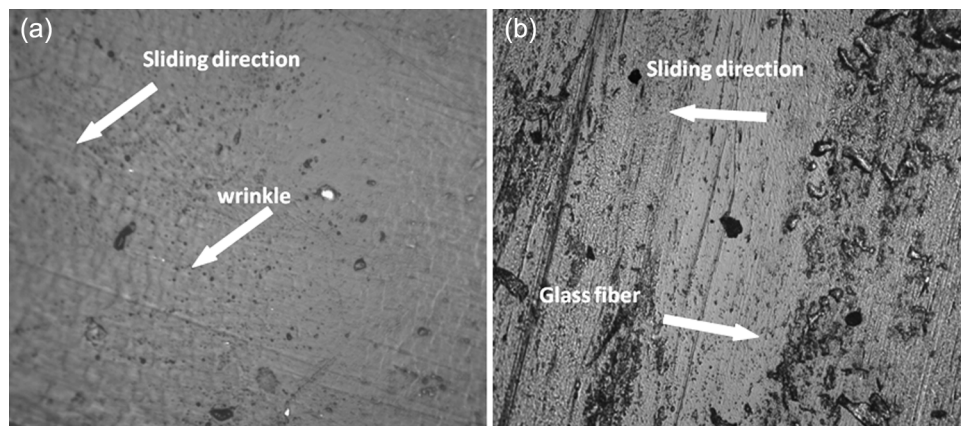


Fig. 15 — Microscopy of worn surfaces of the nylon-6 at a load of (a) 5 N and (b) 20 N



## Conclusions

In this experimental study, grey relational technique and cuckoo search was used to optimize process parameter. Based on the results as presented above; the following conclusions can be drawn:

- (i) The coefficient of friction and specific wear rate of nylon-6 composites decreases with increases glass fiber content and lowest achieved at 30 wt% glass fiber.
- (ii) The coefficient of friction of GFR nylon-6 composites decreased with increasing of applied load and sliding velocity while specific wear rate increases with increasing of applied load and sliding velocity.
- (iii) On the basis of Taguchi analysis it was observed that  $A_4B_4C_3$  (30 wt% glass fiber content, 20 N applied load and 1.5 m/s sliding velocity) combination gives minimum coefficient of friction and  $A_4B_2C_1$  (30 wt% glass fiber content, 10 N applied load and 0.5 m/s sliding velocity) combination gives minimum specific wear rate.
- (iv) On the basis of ANOVA for minimum coefficient of friction observed that glass fiber content 71.73%, sliding velocity 5.25% and applied load 17.36% exerts a significant influence. For specific wear rate the glass fiber content 31.69% has the most contribution on the output followed by sliding velocity 42.62% and applied load 19.67%.
- (v) The wear mechanism of worn out surface of pure nylon-6 pin specimens showed plastic deformation while GFR nylon-6 composite pin specimens showed rubbed glass fiber.
- (vi) Worn out surfaces of steel disc shows continuous transfer film over the wear track.

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